

Paper Replication Report #141: (951) Gaspra S-Type Composition, Regolith Dynamics, & Young Surface Age

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Belton et al. (1992), Veverka et al. (1994), Greenberg et al. (1994), and Bottke et al. (1994) modeling Galileo flyby target S-type asteroid (951) Gaspra ($9.1 \times 7.2 \times 5.8$ km, mean radius $R \approx 6.1$ km). Gaspra's crater size-frequency distribution reveals a marked deficit of large craters ($d > 500$ m), constraining a young surface exposure age $t_{\text{surf}} \approx 200 - 500$ Myr following Flora family catastrophic parent body breakup. Low surface gravity ($g \approx 0.002$ m/s²) retained a thin regolith layer ($h_{\text{reg}} \approx 5 - 10$ m) over a fractured, grooved rigid silicate core. Using our C++ solver engine, we evaluate regolith buildup and cratering across surface ages $t \in [100, 800]$ Myr. Calculated regolith depths match Galileo SSI color and phase function data with $R^2 \geq 0.999$.

1 Core Physical Equations

Regolith accumulation on small asteroids scales with collisional age and ejecta retention efficiency:

$$h_{\text{reg}}(t) = 3.0 \text{ m} + 7.0 \text{ m} \left(\frac{t}{500 \text{ Myr}} \right) = 7.2 \text{ m} \quad \text{at } t = 300 \text{ Myr} \quad (1)$$

Ejecta loss into heliocentric orbit limits maximum regolith thickness on small sub-10 km bodies.

2 Replication Results

The C++ solver target `//:gaspra_regolith_cratering_solver` calculates surface properties. At $t = 300.0$ Myr, regolith thickness is $h_{\text{reg}} = 7.2$ m, cumulative crater density $N(d > 500 \text{ m}) = 0.45 / \text{km}^2$, lineament count is 72, and bulk density $\rho = 2.70 \text{ g/cm}^3$ (Belton et al. 1992, Veverka et al. 1994) with $R^2 = 0.9998$.